

1 Configuration sampling

Device count

234

Sample

3

Device types

Mobile

Desktop

SmartHome

Applications

Mobile

Calendar

Contacts

Messages

...

Desktop

Text editor

Spreadsheet

Browser

...

SmartHome API

Feasibility filter

Sampled configuration

2 Task design

Configuration + device capabilities

User goal (example)

Prepare the guest room...

How should I design this task?

Revise

Task specification

Mobile

Arrival: 20:00

Preferences

Desktop

Lead time: 25 min

Operating rules

Start time

20:00 – 25 min

= 19:35

SmartHome settings

Preferences + rules

AC 24°C Lights 35%

Completion conditions

Schedule correct

Record saved

Confirmation sent

Independent design review

3 Executable instantiation

Approved design

What to initialize?
How to configure?

Resource specification

Configure resources to match task needs

File generation

Spec

Build

Write files

HTML

CSV

JSON

...

XLSX

TXT

...

Application setup

Create initial data

SMS

Email

Contacts

...

Calendar

Notes

...

Initialize

Evaluate

Clean up

Executable task package

4 Validation & Repair

Programmatic checks

Resources accessible

Interfaces available

Targeted execution tests

Mobile

Desktop

SmartHome

Evaluator checks

Expected

Incomplete

What failed?
How to fix it?

Issues

Repair & recheck

Validated task instance

Human review

Goals, roles and dependencies

Device execution

Cross-device feasibility

Revise & recheck

Descriptions, resources and components